

ALLAN ILYASOV

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EDUCATION

St. John's University – Queens, New York

Master of Science, Data Science

May 2026

Bachelor of Science, Computer Science

May 2025

- Dean's List 2021-2025 | Member of the Association for Computing Machinery | EPIC Mentoring Program, Student Mentor

RELEVANT EXPERIENCE

Graduate Research Assistant – St. John's University

September 2025 – Present

Codify AI – Advancing Education Through Generative AI & LLM (Dr. Fazel Keshtkar):

February 2024 – Present

- Product manager & lead architect of codifyai.org, an AI-powered programming tutor using LangChain agents, Claude Haiku 4.5 via AWS Bedrock, and RAG for streaming responses, and context-aware tutoring in Java, Python & SQL.
- Designed production-ready AWS infrastructure (Elastic Beanstalk, DynamoDB, Route 53, HTTPS) with Docker containerization, supporting scalable REST API architecture.
- Built React-based adaptive learning platform with persistent state management, real-time progress tracking.
- Earned “Commercializable Project” award at the 2025 St. John's University's Student Research Conference.
- Presented at the Middle States Commission on Higher Education (MSCHE) Conference. stjohns.edu/news-media

ADDITIONAL EXPERIENCE

Manager – Uncle Edik's Pickles

December 2019 – Present

- Grew a home-based pickle start-up and supported its growth into a brick-and-mortar factory and national brand.
- Trained & led a team of 5 employees in production of thousands of jars per week of 9 different pickled products.
- Managed sales to customers, wholesale delivery to supermarkets, and inventory replenishment from Restaurant Depot.
- Designed branded merchandise, and produced viral marketing content, generating millions of views.
- Ensure compliance with NYS Food Safety regulations & maintained quality control with daily production checklists.

SKILLS SUMMARY

- **Programming Languages:** Python, Java, SQL, R, JavaScript, Swift, TypeScript, C++, PHP
- **Frameworks & Libraries:** React, Next.js, Django, Flask, Spring Boot, Node.js, Bootstrap, LangChain, LangGraph, REST APIs, Pandas, NumPy, Scikit-learn, TensorFlow/Keras, PyTorch, Spark, Beautiful Soup, NLTK, Spacy
- **Cloud & DevOps:** AWS, Docker, DynamoDB, Elastic Beanstalk, Lambda, API Gateway
- **Language Skills:** English and Russian native speaker | Intermediate proficiency in Spanish

PROJECTS

Slider - AI-Powered PowerPoint Add-in | github.com/Slider-ai/hackathon2026

February 2026

- Collaborated on a PowerPoint add-in with a chat interface that generates slides from prompts using GPT-4o.
- Built chat UI & conversation flow with intent parsing to route prompts across generate, summarize, & research modes.
- Integrated Tavily live web search and URL fetching to pull real-time content from articles and webpages into slides.

Bluetooth Optimized Morning Bell – Alarm Clock & IOS App | github.com/allanilya/B.O.M.B.

January 2026

- Built an alarm clock using an ESP32 and e-ink display to reduce nighttime light exposure and eliminate phone dependency during wake-up, featuring a physical snooze and dismiss button and an I2S DAC amplifier with speaker output.
- Designed a BLE-based iOS companion app for initial setup and configuration, enabling alarm scheduling, brightness control, and custom sound uploads while keeping daily alarm operation fully device-local.

Time Series Forecasting for Financial Markets | github.com/allanilya/Stock-analytics

October 2025 – November 2025

- Created ARIMA, LSTM, & GRU forecasting models across AAPL, NVDA, & LYFT, reducing RMSE by up to 82%.
- Implemented grid search across 36 ARIMA configurations and built 2-layer LSTM/GRU architectures with dropout regularization utilizing Python, TensorFlow, and Statsmodels.
- Conducted stationarity analysis via ADF tests and automated extraction of 1,458+ trading days using yfinance api.

Multimodal Video Analysis | github.com/allanilya/Multimodal-Video-Analysis

June – July 2025

- Developed full-stack AI video analysis platform with Flask/Next.js featuring CLIP-based visual search, transcript search, and Gemini 2.5 Flash multimodal chat with conversational memory.
- Engineered async parallel processing pipeline achieving <3s response time via background frame extraction.

Diabetes Risk Prediction | github.com/allanilya/Diabetes-Prediction.

April 2025

- Investigated whether lifestyle or demographic factors are greater predictors of diabetes using 230K+ patient records, discovered lifestyle to be more significant.
- Applied data wrangling (imputation, encoding, outlier detection) and addressed class imbalance with SMOTE.
- Developed Random Forest, Logistic Regression & KNN models with 5-fold cross-validation to predict diabetes risk.

Work Eligibility: US Citizen